What Do Vision-Language Models Really See? Ordinary-Bench: Evaluating Spatial Belief via Controlled Probing and Scene Reconstruction

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Abstract

The deeper function of vision is not merely to output labels, but to recover a stable and actionable model of the external world from incomplete and noisy visual input. Inspired by psychophysics and systems neuroscience, we treat a vision-language model (VLM) as a black-box observer and probe its *externalized scene belief* through dense ordinal spatial questions. This is motivated by a simple observation from ordinal embedding: if sufficiently many ordinal relations are correct, a scene can be approximately reconstructed up to a similarity transform; asking a VLM enough QRR and TRR questions is therefore a way of testing what kind of scene the model effectively "has in mind." Our current experiments on 540 completed scenes and over 200K QA pairs show that scene-level spatial understanding remains weak: single-view QRR reaches only about 70% accuracy, multi-view QRR drops toward chance, TRR stays near chance, and neither supervised fine-tuning nor reinforcement learning on these relational QA pairs yields substantial gains. Ordinary-Bench thus shifts evaluation from "how many spatial questions did the model answer correctly?" to "does the model support a coherent, reconstructable, and alignable spatial world model at all?" and opens a path toward vectorized scene reconstruction, localization, and structured scene memory from ordinal relations.

1 Introduction

The purpose of vision is not exhausted by recognition. For humans, visual perception supports localization, action, memory, navigation, and imagination because it helps construct a structured model of the external world from partial retinal evidence. In psychophysics and systems neuroscience, this is precisely why researchers do not stop at isolated accuracy scores: they rely on controlled stimuli, repeated probing, perturbations, and cross-condition comparisons to infer what kind of representation an observer actually maintains Kriegeskorte et al. [2008], Yamins and DiCarlo [2016]. If we want to understand what a VLM really sees, we should adopt the same posture. A model should not be treated merely as a question-answering device; it should be treated as an observer whose behavior may or may not imply a coherent scene representation.

Current VLM evaluation falls short of this standard. Many benchmarks reduce spatial competence to isolated judgments such as "left of," "closer than," or "in front of." A correct answer to one question, however, does not imply that the model has formed a global scene. Even many individually correct answers may be mutually incompatible once interpreted jointly. A model may appear competent at local distance comparison while failing to assemble those judgments into a single consistent layout; it may produce plausible directional answers by leaning on language priors or surface heuristics, yet those answers may immediately collapse when placed into a unified coordinate system. What most existing spatial benchmarks measure, therefore, is often local relational answering rather than reconstructable spatial belief.

Our motivation comes from a basic fact in ordinal embedding and scene reconstruction: when ordinal relations are sufficiently dense and sufficiently accurate, scene geometry can be approximately recovered up to similarity transforms Agarwal et al. [2007], Ma et al. [2018], Dokmanić et al. [2015]. This suggests a new way to evaluate VLMs. If we ask a model enough relative-distance and relative-direction questions, the resulting answer set becomes a form of behavioral tomography. QRR and TRR are not just tasks; they are probes of the model's externalized scene belief. QRR tests whether the model preserves the relative near-far structure among objects. TRR tests whether it maintains a coherent reference frame for direction. If the induced constraint set supports a reconstruction close to the ground-truth scene, then the model likely maintains a coherent scene-level representation. If not, then local success is better explained as a patchwork of priors, templates, or weak independent heuristics.

This is the motivation behind **Ordinary-Bench**. Rather than adding yet another spatial QA dataset, we treat a VLM as an observer to be behaviorally reverse-engineered. We generate controlled 3D-rendered scenes, probe them densely with QRR (quaternary relative relations) and TRR (ternary clock-direction relations), and then interpret the model's answers as ordinal and directional constraints whose consistency and reconstructability can be tested. The use of controlled synthetic scenes is deliberate. For the question "what scene does the model actually externalize?", exact geometric ground truth, controlled complexity, and repeatable perturbations are more valuable than raw ecological realism Johnson et al. [2017], Yi et al. [2020].

The empirical picture that motivates this paper is already clear. On the 540 scenes currently completed, single-view QRR reaches only about 70% accuracy. More strikingly, adding multiple views does not improve this capability; instead, multi-view QRR degrades toward chance. TRR remains near chance throughout. This means that current VLMs may retain a weak and fragile notion of relative distance, but they still fail to form a reliable scene-level directional representation. Even more importantly, supervised fine-tuning or reinforcement learning on more than 200K relational QA pairs does not yield a meaningful improvement. This negative result matters: simply injecting more relational supervision into a model does not automatically produce a geometrically coherent scene belief. If evaluation stops at question-level accuracy, this structural failure is easy to miss.

From an application perspective, the stakes go beyond spatial QA. If a VLM can support stable scene reconstruction from ordinal relations, then it effectively exposes a vectorized scene representation that is only weakly coupled to any particular metric scale. Once the ordinal structure is stable, absolute scales can be aligned afterward using task-specific priors, empirical calibration, or ground-truth anchors. This creates a path toward scene reconstruction, localization, retrieval, and structured scene memory from relational queries alone. Ordinary-Bench is therefore not only about whether a model can answer spatial questions, but whether it has reached the more fundamental ability to recover a manipulable spatial structure from vision.

Our contributions are three-fold. First, we introduce a psychophysics-inspired evaluation framework for VLM spatial understanding, using controlled 3D scenes and dense QRR/TRR probing to study the model as a behaviorally observable observer. Second, we bring the idea of reconstructability from ordinal embedding into multimodal evaluation: enough relational answers should, in principle, reveal whether the model holds a coherent scene belief. Third, we present a set of already informative negative findings: limited single-view QRR success does not translate into stable spatial belief, multi-view input can collapse rather than improve performance, TRR remains near chance, and large-scale relational fine-tuning or RL does not substantially change this picture. Taken together, these results push the evaluation question from "can a VLM answer spatial questions?" to the stricter question: *does the model actually construct a coherent, reconstructable world from visual input?*

2 Related Work

2.1 Psychophysics, systems neuroscience, and behavioral reverse engineering

Our methodology is inspired by the broader logic of psychophysics and systems neuroscience, where the goal is not merely to score an observer on isolated stimuli, but to infer the structure of its latent representation through controlled probes and perturbations. Representational similarity analysis emphasizes that structure can be inferred behaviorally even when the underlying mechanism is not directly accessible Kriegeskorte et al. [2008]. Goal-driven modeling in visual neuroscience similarly argues that what matters is not one task score in isolation, but whether a model reproduces systematic representational structure and robust invariances Yamins and DiCarlo [2016]. We adopt this stance for VLMs: the paper does not claim to decode the model's internal neural state directly, but instead studies the externalized scene belief implied by its answers.

2.2 Controlled synthetic worlds for visual reasoning

CLEVR and CLEVRER established the scientific value of controlled synthetic benchmarks when the aim is to study reasoning rather than ecological realism alone Johnson et al. [2017], Yi et al. [2020]. By controlling object attributes, compositional structure, and event dynamics, these benchmarks make it possible to localize failure modes much more precisely than open-world datasets typically allow. Ordinary-Bench inherits this experimental logic, but focuses specifically on whether a model's answers imply a reconstructable spatial world. For this purpose, exact geometry, controllable complexity, and repeatable visual perturbations are essential.

2.3 Spatial benchmarks for VLMs

Recent work has shown that spatial understanding remains a persistent weakness for VLMs. SpatialVLM targets spatial reasoning more directly Chen et al. [2024a], while VSR and What'sUp document failures on visual spatial relations and directional understanding Liu et al. [2023], Kamath et al. [2023]. These benchmarks are important, but they still largely frame the problem as one of question-level correctness. They ask whether the model can solve a spatial item, not whether all of its answers together define a coherent scene. Ordinary-Bench differs by taking scene reconstructability rather than local correctness as the central criterion.

2.4 Language priors, hallucination, and evaluation mismatch

Another line of work argues that benchmark success can be inflated by language priors or evaluation artifacts. BLINK highlights the gap between seeing and perceiving in multimodal models Fu et al. [2024]; MMStar questions whether current VLM evaluation pipelines measure the intended abilities at all Chen et al. [2024b]; POPE studies object hallucination Li et al. [2023]; and VLInd-Bench explicitly measures the contamination of visual answers by language priors Lee et al. [2025]. Ordinary-Bench is aligned with this critique, but pushes it one step further. The issue is not only whether a model can answer correctly without vision, but whether its answers support any scene that could actually be reconstructed.

2.5 Selective answering and reliability

Reliable multimodal evaluation should also account for uncertainty and abstention. Reliable VQA argues that abstaining is often better than confidently answering incorrectly Whitehead et al. [2022]; Selectively Answering Visual Questions and later consistency-based selective VQA frameworks extend this argument to black-box VLMs Eisenschlos et al. [2024], Khan and Fu [2024]. This perspective is highly relevant here because once answers are interpreted as geometric constraints, a confidently wrong answer is not just a single mistake: it can corrupt the entire reconstructed scene. For this reason, Ordinary-Bench is naturally compatible with selective-answering and consistency-oriented protocols.

2.6 Ordinal embedding and geometric realizability

The immediate theoretical basis of our evaluation lies in ordinal embedding and geometric realizability. Generalized non-metric multidimensional scaling and later robust ordinal embedding work show that relative comparisons can recover geometric structure even without absolute metric values Agarwal et al. [2007], Ma et al. [2018]. Euclidean distance matrix theory reminds us, however, that a set of ordinal constraints is not automatically geometrically realizable Dokmanić et al. [2015]. For directional relations, bearing and angle rigidity provide the right language for understanding uniqueness, mirror ambiguity, and the realizability of directional constraints Zhao and Zelazo [2019], Chen et al. [2021]. Ordinary-Bench does not claim a new general-purpose reconstruction algorithm; instead, it turns these ideas into an evaluation principle for VLMs: if enough accurate ordinal relations can reconstruct a scene, then dense relational probing can reveal whether a model possesses a coherent scene belief at all.

2.7 How our work differs

Existing work typically concentrates on one of three questions: whether VLMs can answer spatial questions, whether visual evaluation is contaminated by language priors, or whether ordinal constraints can reconstruct geometry. Ordinary-Bench combines all three. It adopts a psychophysics-inspired black-box observer view, uses controlled synthetic scenes, and treats geometric reconstructability as the end-point of evaluation. In this sense, the benchmark is not simply another spatial QA dataset; it proposes a stricter criterion for vision-language understanding: a model should count as spatially competent only if its many local answers jointly support a coherent, reconstructable, and alignable scene.

3 Ordinary-Bench: Data and Protocol

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4 Scene Belief Reconstruction

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5 Experiments

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6 Discussion and Outlook

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